Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[english]babel [T1]fontenc lmodern,mathrsfs xparse [inline,shortlabels]enumitem topsep=2pt,itemsep=2pt,parsep=0pt
[dvipsnames]xcolor [utf8]inputenc [a4paper,top=0.5in,bottom=0.2in,left=0.5in,right=0.5in,footskip=0.3in,includefoot]ge
[most]tcolorbox minted minted [colorlinks]hyperref
pythoncodebox/.style= enhanced jigsaw,breakable, colback=gray!10,colframe=gray!20!black, bo-
xrule=1pt,top=2pt,bottom=2pt,left=2pt,right=2pt, sharp corners,before skip=10pt,after skip=10pt,
attach boxed title to top left, boxed title style=empty, top=0pt,bottom=0pt,left=2pt,right=2pt, in-
terior code=[fill=tcbcolframe] (frame.south west) –([yshift=-4pt]frame.north west) to[out=90,in=180]
([xshift=4pt]frame.north west) –([xshift=-8pt]frame.north east) to[out=0,in=180] ([xshift=16pt]frame.south
east) –cycle; , title=1, fonttitle=, pythoncodebox/.default=, pythoncodebox*/.style= enhanced ji-
gsaw,breakable, colback=gray!10,coltitle=gray!20!black,colbacktitle=tcbcolback, frame hidden, top=2pt,bottom=2pt,left
sharp corners,before skip=10pt,after skip=10pt, attach boxed title to top text left=yshift=-1mm,
boxed title style=empty, top=0pt,bottom=0pt,left=2pt,right=2pt, interior code=[fill=tcbcolback]
(interior.south west) –([yshift=-4pt]interior.north west) to[out=90,in=180] ([xshift=4pt]interior.north
west) –([xshift=-8pt]interior.north east) to[out=0,in=180] ([xshift=16pt]interior.south east) –cycle;
, title=1, fonttitle=, pythoncodebox*/.default=,
pythonbox[1][]pythoncodebox=1 pythonbox*[1][]pythoncodebox*=1
pythonautogobble,breaklines,mathescape
python !O !D() !G listing engine=minted, listing only, pythoncodebox=1, minted langua-
ge=python, minted options/.expanded= autogobble,breaklines,mathescape, 2 , 3
python* !O !D() !G listing engine=minted, listing only, pythoncodebox*=1, minted langua-
ge=python, minted options/.expanded= autogobble,breaklines,mathescape, 2 , 3
verbbox !O listing engine=minted, minted language=latex, boxrule=1pt,sidebyside,skin=bicolor,
colback=gray!10,colbacklower=white,valign=center, top=2pt,bottom=2pt,left=2pt,right=2pt, 1
Typing Python in LaTeX Senan Sekhon April 10, 2023

This template provides environments for rendering aesthetic, professional-looking Python code in LaTeX.

The code is rendered using the `minted` package, which is based on Pygments. The boxes surrounding them are created using the `tcolorbox` package. Note that this package must be loaded with the `minted` library, e.g. by adding latexminted to the preamble, for these to work.

The `pythonbox` environment creates a box for Python code (just the box, not the code). The starred version `pythonbox*` creates a "soft" box with no border.

This is a `pythonbox`.

[The title] This is a `pythonbox` with a title.

This is a soft `pythonbox`.

[The title] This is a soft `pythonbox` with a title.

The `python` environment creates Python code inside a `pythonbox`. In this environment, you can directly enter Python code and `minted` will color the text accordingly. Similarly, the starred version `python*` has no border.

def next$_t wo(x) : lst = [x + if oriinrange(3)]returnlst$

[The function] def next$_t wo(x) : lst = [x + if oriinrange(3)]returnlst$

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The `python` and `python*` boxes have three optional arguments:

- The first argument, delimited by brackets `[ ]`, specifies the title.
- The second argument, delimited by parentheses `( )`, specifies the options for the `minted` environment, i.e. the code itself. These are handled by the `minted` package.
- The third argument, delimited by braces `{ }`, specifies the options for the box containing the code. These are handled by the `tcolorbox` package.

[sidebyside=false] [How to calculate a factorial](style=one-dark)colback=gray!40!black,colframe=blue def factorial(n): product=1 Start with 1 for k in range(2,n+1): For each $k = 2, 3, \ldots, n$, product*=k Multiply by k return product

In the above example, the option `style=one-dark` is an option for the *code*, while the options `colback=gray!40!black` and `colframe=blue` are options for the *box*. Note that the option `style` only specifies the style for the code itself, it does not change the background color — this has to be done separately by specifying the `tcolorbox` option `colback`[1].

See `https://pygments.org/styles/` for a list of available styles. You can also use other `tcolorbox` environments[2].

The pre-defined options for the code are `autogobble` (this removes any common indentation from all lines of code), `breaklines` (to allow lines that are too long to be broken) and `mathescape` (to allow LaTeX math to be rendered in code comments). See the documentation for the `minted` package for a list of available options.

[1] This *cannot* be achieved by specifying the `minted` option `bgcolor` — doing this would only create a colored box around the code, rather than coloring the entire interior of the `tcolorbox`.

[2] For example, the `filingbox`, `railingbox` and `flagbox` environments from this template.

def f(x): This function takes a number x and returns x^2 return x**2

You can also typeset Python code by itself (without the box) using the `pythoncode` environment. This is based on the latex command from the `minted` package, and contains the same pre-defined options as `python`.

def least$_s quare_a bove(x) : return 0 if x < 0 else (int(x * *0.5) + 1) * *2$

You can overwrite these options or provide additional options by using `pythoncode*`, but it is better to use the standard `minted` environment: latex[⟨*options*⟩latex]python ... .

style=manni def least$_s quare_a bove(x) : return 0 if x < 0 else (int(x * *0.5) + 1) * *2$

Finally, note that LaTeX does not *run* the Python code, it only typesets it. If you want to run the code, you can paste it into TutorialsPoint:

`https://www.tutorialspoint.com/online_python_compiler.php`

Or you can use any IDE, such as PyCharm or Visual Studio Code.